A WEB of Sharing Trust–Responsibility Toward a Trustless Future vol.2

Governance Stress-Testing and Ethical Evaluation

Second Edition — Version D

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Distribution Statement

This publication is released as a **Public Review Draft (Standards**

Exploration). It may be used for:

- academic research and comparative analysis,
- policy development and regulatory consultation,
- standards exploration and working-draft discussion, and
- evaluation of approaches to AI governance that move beyond static, documentation-based compliance toward **engineered, runtime trust mechanisms**.
- Feedback is invited to support refinement prior to subsequent editions.

This document does **not** constitute regulatory guidance, certification, or legal advice. It is intended to support informed review, discussion, and refinement of governance architectures, assurance models, and scoring methodologies for AI systems.

Second Edition Note — Version D Governance Stress-Testing and Ethical Evaluation

Canonical paper:
APOD-TR-005_Vol2_Version-D (PDF)

Informative companion artifacts are published with the GitHub release:

v1d-Governance Stress Testing and Ethical Evaluation

https://github.com/Access-PoD/access-pod-artifacts/releases/tag/v1d

Overview

Version D extends the *A WEB of Sharing Trust–Responsibility Toward a Trustless Future* platform by introducing **structured governance stress-testing and ethical evaluation**.

While earlier editions establish governance authority (Version A), demonstrate execution-level feasibility (Version B), and formalize onboarding and containment (Version C), **Version D addresses a critical unresolved question**:

How does governance behave when it is deliberately stressed before harm occurs?

Version D treats governance itself as a **system under load**, and ethics as a **decision capability exercised under uncertainty**, rather than as a checklist, alignment label, or certification claim.

What This Release Introduces

Version D introduces a disciplined, non-exploitative approach to governance stress-testing, including:

- Defined **governance stress categories**, such as:
  - authority stress,
  - delegation stress,
  - telemetry stress,
  - incentive stress, and
  - ethical ambiguity stress.
- **Non-exploitative stress-testing methods**, grounded in simulation, walkthroughs, and revocation rehearsal rather than adversarial hacking or red-team activity.
- A clear treatment of **ethics under governance stress**, focusing on responsibility, restraint, escalation, and non-decision under uncertainty.
- Formal rules for **how stress-testing evidence may constrain trust and assurance**, without modifying or extending Compliance Star Certificate (CSC) semantics.

- Explicit guardrails preventing **stress-testing-as-certification**, benchmarking, or marketing misuse.

Stress-testing failures are treated as **valuable governance evidence**, not defects to be concealed or reframed.

What Version D Does *Not* Do

Version D does **not**:

- certify systems or approve deployment,
- introduce new assurance or scoring logic,
- validate alignment claims,
- function as a security red-team or exploit program,
- or act as a market-access or compliance gate.

Stress-testing under Version D **constrains trust**; it never elevates it.

Relationship to Prior Versions

- **Version A** remains the *authoritative reference* for governance authority, terminology, assurance semantics, CSC logic, and conformance meaning.
- **Version B** remains an *illustrative, architecture-specific extension*, demonstrating execution-level feasibility without redefining governance authority.
- **Version C** remains the *onboarding and containment* reference, formalizing governed entry into runtime operation.
- **Version D** adds **governance and ethical stress-testing**, without altering authority, scoring, or certification semantics.

Where differences or ambiguity arise, **Version A prevails**.

All stable, published papers in this series are made available for public download via the Policy of Developments website:

https://policyofdevelopments.org/whitepaper/vol2-a-web-of-sharing-trust-responsibility-toward-a-trustless-future

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Relationship to the v1.0-Workshop

Effective use of Version D presumes the **human governance capability** developed through the **v1.0-workshop — Designing Governance-Ready AI Outputs**.

The workshop is represented in two complementary forms. The public workshop repository contains Markdown-based materials intended for inspection, versioning, and standards-adjacent review. The workshop information pack published on the website provides a structured overview of the workshop's purpose, scope, and learning outcomes for prospective participants.

These serve different audiences and functions: the repository supports technical and governance transparency, while the website materials support orientation and engagement without conveying governance authority or certification.

The workshop develops disciplined professional practice in:

- articulating governance intent and scope,
- identifying non-delegable judgment,
- reasoning under uncertainty,
- documenting defensible decisions, and
- exercising escalation and non-decision responsibly.

Without this human capability substrate, governance stress-testing risks becoming abstract or performative. With it, Version D becomes **practical, accountable, and decision-relevant**.

Informative reference:

Workshop repository (v1.0-workshop):
https://github.com/Access-PoD/access-pod-artifacts/releases/tag/v1.0-workshop

Additional information about the Designing Governance-Ready AI Outputs workshop, including an overview and participant information pack, is available at:

https://policyofdevelopments.org/workshop

When referencing this release, it should be cited as a facilitated professional learning framework focused on governance reasoning, evidence discipline, and accountable AI reporting.

It should not be cited as regulatory guidance, certification criteria, or a compliance determination.

Intended Audience

This release is intended for:

- regulators and oversight bodies exploring runtime governance resilience,
- standards bodies evaluating governance robustness under stress,
- organizations prior to scale-up or scope expansion,
- boards and risk committees accountable for trust claims.

It is **not intended** for public benchmarking, marketing claims, or certification-by-association.

Appendix — Validation Results (Informative)

The Version D companion artifacts in - v1d-Governance Stress Testing and Ethical Evaluation: include machine-readable JSON files intended to support **inspectable, repeatable governance stress-testing**.

Prior to release, the following artifacts were structurally validated using the provided validate_jsonschema.py script:

- v1d_scenarios.index.json
  validated against v1d_scenarios.schema.json
- apod.v1d.stress_evidence_record.example.json
  validated against apod.v1d.stress_evidence_record.schema.json

Validation confirms that these files **conform to the published schemas** and contain all required fields with permitted values.

Validation is **structural only**. It does not assess governance quality, ethical sufficiency, trust posture, or compliance, and it does not confer approval, certification, or endorsement.

The validation process is provided to support transparency, inspection, and standards-adjacent review.

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Executive Summary — Version D

Governance Stress-Testing and Ethical Evaluation

Status: Planned (Standards Exploration)

As AI systems become adaptive, compositional, and embedded in decision-critical environments, governance mechanisms must be evaluated not only under nominal conditions, but under stress. Frameworks that appear robust in documentation or steady-state operation may fail when authority is contested, evidence degrades, incentives conflict, or ethical ambiguity emerges.

Volume 2 — Version D introduces **structured, non-exploitative governance stress-testing** as a first-class governance capability. Its purpose is to surface governance failure modes *before harm occurs* by examining how authority, escalation, telemetry, and ethical judgment behave under pressure.

Version D does not introduce new governance principles, assurance semantics, or certification logic. It is explicitly subordinate to **Version A**, which remains authoritative for governance authority, terminology, assurance semantics, and Compliance Star Certificate (CSC) interpretation. It is complementary to **Version B**, which demonstrates execution-level feasibility, and **Version C**, which formalizes onboarding and containment as governed transitions.

The core contribution of Version D is to treat **governance itself as a system under load**. Rather than testing model performance or exploiting technical vulnerabilities, Version D focuses on:

- authority stress (who decides when governance becomes inconvenient),
- delegation stress (how accountability survives compositional systems),
- telemetry stress (how governance responds to incomplete or contested evidence),
- incentive stress (when governance conflicts with organizational goals), and
- ethical ambiguity stress (when harm is plausible but not provable).

Ethics, within Version D, is not framed as a principles checklist or alignment label. It is treated as a **governance capability exercised under uncertainty**, revealed through decisions, restraint, escalation, and accountability when evidence is incomplete or outcomes are contested.

Governance stress-testing methods defined in Version D are deliberately constrained. They are simulation-based, reversible, human-supervised, and non-exploitative. They generate governance evidence - such as clarified boundaries, decision rationales, and identified failure modes - but do not produce approvals, certifications, or readiness determinations.

Stress-testing evidence interacts with assurance **only by constraint**. It may narrow scope, justify provisional assurance, or rehearse revocation readiness, but it cannot elevate trust signals or justify deployment. Trust that cannot degrade under stress is not trustworthy.

Version D is intended for regulators, standards bodies, system operators, boards, and risk committees at **governance inflection points** - such as prior to scale-up, scope expansion, or integration into higher-impact contexts. It is not intended for public benchmarking, marketing claims, or certification-by-association.

Effective use of Version D presumes the human governance capabilities developed through the **v1.0-workshop**, which provides the reasoning discipline required to articulate boundaries, own accountability, and act under uncertainty. Without this human substrate, governance stress-testing risks becoming abstract or performative.

Taken together, Versions A–D and the workshop form a coherent platform:

- **Version A** defines governance authority and assurance semantics.
- **Version B** demonstrates execution-level feasibility.
- **Version C** constrains onboarding and early operation under containment.
- **Version D** stress-tests governance and ethical boundaries before harm.
- **v1.0-workshop** develops the human capability required to exercise all of the above responsibly.

Version D reinforces a central principle of the platform:

Trust in AI systems must be engineered, observed, constrained, and when necessary withdrawn before impact occurs.

Chapter 1 — Why Governance Must Be Stress-Tested

1.1 The Limits of Nominal Governance

AI governance frameworks are typically evaluated under **nominal conditions**:
stable assumptions, cooperative actors, intact telemetry, and aligned incentives.

Under such conditions, governance mechanisms often appear sufficient. Policies are documented, controls are declared, escalation paths exist on paper, and ethical principles are articulated clearly.

However, **nominal conditions are not where governance fails**.

Governance failures emerge when:

- incentives conflict with stated controls,
- uncertainty increases beyond expected bounds,
- accountability becomes inconvenient,
- evidence is incomplete or contested, or
- human intervention carries cost or delay.

Existing governance approaches rarely test these conditions explicitly. As a result, they risk mistaking *designed intent* for *operational resilience*.

Version D addresses this gap by treating governance not as a static construct, but as a system that must **withstand stress** before it can be trusted.

1.2 Governance as a System Under Load

In mature engineering disciplines, systems are not evaluated solely by design documentation or nominal performance. They are **stress-tested** to observe behavior under adverse or extreme conditions.

Examples include:

- financial stress testing to assess capital adequacy under market shock,
- safety-case challenge scenarios in aviation and nuclear systems,
- disaster recovery drills in critical infrastructure.

These practices do not assume failure is exceptional. They assume failure is **informative**.

AI governance, by contrast, is often evaluated only by:

- whether policies exist,
- whether documentation is complete,
- whether principles are declared.

This creates a structural blind spot:
governance is rarely observed while it is being strained.

Version D treats governance mechanisms - policies, escalation logic, human-in-the-loop controls, assurance signals - as **systems that must perform under load**, not merely exist in specification.

1.3 Why Ethics Cannot Be Evaluated at Design Time Alone

Ethical evaluation of AI systems is frequently conducted as a **principles exercise**:

- fairness,
- accountability,
- transparency,
- human oversight.

While such principles are necessary, they are insufficient to predict behavior when:

- trade-offs become explicit,
- harms are plausible but not yet realized,
- benefits accrue unevenly,
- or ethical action conflicts with organizational pressure.

Ethics, in practice, is not exercised in abstract alignment statements.
It is exercised in **decisions under constraint**.

Version D therefore rejects the framing of ethics as a checklist or value declaration. Instead, ethics is treated as a **governance stress condition** revealed through:

- contested decisions,
- ambiguous evidence,
- delayed escalation,
- or pressure to proceed without certainty.

Ethical failure modes often emerge **only when governance is stressed**, not when it is described.

1.4 The False Safety of "Compliant Until Proven Otherwise"

Many AI governance regimes implicitly assume:

A system is acceptable until evidence of harm appears.

This assumption places governance in a **reactive posture**, where failure must occur before controls are questioned.

Version A explicitly replaced this logic with **continuous, evidence-based assurance**.
Version C further constrained onboarding by requiring containment before alignment.

Version D extends this trajectory by asking a harder question:

What happens to governance controls before harm, when stress is deliberately applied?

Governance stress-testing does not wait for incidents.
It explores **where incidents would emerge if conditions worsened**, and whether governance mechanisms respond proportionately.

This approach treats stress-testing as a **preventive governance activity**, not a forensic one.

1.5 Why Stress-Testing Governance Is Not Red-Teaming

It is important to distinguish Version D from:

- security red-teaming,
- adversarial hacking,
- model jailbreak exercises,
- exploit disclosure programs.

Those activities focus on **breaking systems**.

Version D focuses on **breaking governance assumptions**.

Stress-testing in this context does not seek to:

- extract restricted outputs,
- bypass technical safeguards,
- expose vulnerabilities for exploitation.

Instead, it seeks to surface:

- where authority becomes ambiguous,
- where escalation stalls,
- where telemetry fails to inform decision-making,
- where ethical responsibility becomes diffused.

Stress-testing governance is therefore **non-exploitative by design** and operates within explicitly declared ethical rules of engagement.

1.6 Stress-Testing as a Precondition for Trust

Trust in AI systems is often treated as an outcome of:

- compliance claims,
- certification,
- or maturity labeling.

Version A reframed trust as a **runtime, evidence-derived property**.
Version C constrained trust during onboarding.

Version D adds a final discipline:

Trust must survive stress before it can scale.

A governance system that performs well only when conditions are favorable is not trustworthy - it is fragile.

By introducing structured governance stress-testing, Version D enables:

- earlier identification of failure modes,
- safer refinement of controls,
- clearer ethical boundary articulation,
- and more credible assurance constraints.

Stress-testing does not weaken governance credibility.
It **strengthens it by removing false confidence**.

1.7 Chapter Summary

This chapter establishes why governance must be stress-tested as a system, not merely described as a framework.

Key points:

- Governance failures emerge under pressure, not under nominal conditions.
- Ethics is revealed through stressed decision-making, not principle declaration.
- Stress-testing governance is distinct from red-teaming or exploitation.
- Trust that cannot withstand stress should not be asserted at scale.

Version D therefore treats stress-testing as a **core governance capability**, not an optional exercise or a post-incident response.

Chapter 2 — Types of Governance Stress

2.1 Why Governance Stress Must Be Categorized

Governance stress does not manifest as a single condition. It emerges through **distinct pressure modes**, each revealing different failure patterns in authority, accountability, and ethical control.

Without categorization, stress-testing risks becoming anecdotal or reactive. With categorization, stress-testing becomes a **repeatable governance discipline**, enabling:

- systematic evaluation across systems and sectors,
- comparable interpretation of outcomes,
- and targeted refinement of controls.

Version D therefore defines a set of **governance stress categories**. These categories do not correspond to technical faults or security exploits; they correspond to **decision pressure points** where governance assumptions are most likely to fail.

2.2 Authority Stress

Who decides when governance becomes inconvenient

Authority stress occurs when governance decision rights are unclear, contested, or bypassed under pressure.

Typical triggers include:

- urgent operational deadlines,
- conflicting internal stakeholders,
- external commercial or political pressure,
- escalation fatigue or perceived “over-governance.”

Under authority stress, governance fails not because controls are absent, but because:

- decision ownership becomes ambiguous,
- escalation is deferred or diluted,
- or authority is informally reassigned without traceability.

Stress-testing authority examines whether:

- governance authority remains explicit under time pressure,
- escalation pathways are exercised rather than bypassed,

- and decision responsibility remains human, traceable, and defensible.

Authority stress reveals whether governance is **structural** or merely **procedural**.

2.3 Delegation Stress

What happens when execution is outsourced or composed

Modern AI systems routinely delegate execution to:

- external foundation models,
- third-party APIs,
- tools and agents outside organizational control.

Delegation stress arises when governance assumptions do not scale across these boundaries.

Common failure modes include:

- implicit transfer of responsibility to external components,
- loss of visibility into downstream actions,
- assumptions that third-party alignment substitutes for local governance.

Stress-testing delegation examines whether:

- governance authority remains intact across delegation boundaries,
- telemetry captures delegated actions with sufficient attribution,
- and accountability does not diffuse as systems become more compositional.

Delegation stress tests whether governance survives **ecosystem complexity**, not just internal operation.

2.4 Telemetry Stress

What happens when evidence is incomplete, delayed, or contested

Version A establishes telemetry as the foundation of assurance.
Version D asks what happens when telemetry is **imperfect**.

Telemetry stress occurs when:

- evidence arrives late,

- signals conflict,
- coverage is partial,
- or telemetry is technically present but operationally unusable.

Governance failure under telemetry stress often takes the form of:

- proceeding “temporarily” without evidence,
- treating absence of evidence as evidence of absence,
- or relying on narrative explanations to compensate for signal gaps.

Stress-testing telemetry examines whether:

- governance decisions degrade gracefully when evidence weakens,
- assurance signals are constrained rather than asserted,
- and escalation or pause mechanisms activate appropriately.

Telemetry stress distinguishes **evidence-based governance** from evidence-optional governance.

2.5 Incentive Stress

When governance conflicts with organizational goals

Governance systems operate within organizations that also pursue:

- growth,
- efficiency,
- competitive advantage,
- reputational positioning.

Incentive stress occurs when these goals conflict with governance constraints.

Common scenarios include:

- pressure to deploy despite unresolved risk,
- reluctance to trigger revocation or downgrade signals,
- framing governance outcomes to preserve optics.

Stress-testing incentives examines whether:

- governance controls withstand commercial or reputational pressure,
- assurance mechanisms remain neutral under success narratives,
- and ethical considerations are not subordinated to short-term gain.

Incentive stress reveals whether governance is **independent** or **instrumentalized**.

2.6 Ethical Ambiguity Stress

When harm is plausible but not provable

Not all ethical risks are immediate, measurable, or catastrophic.

Ethical ambiguity stress arises when:

- potential harm is diffuse,
- impacts are long-term or indirect,
- or affected parties are not immediately visible.

Examples include:

- subtle bias amplification,
- erosion of user autonomy,
- normalization of opaque decision-making.

Under ethical ambiguity stress, governance often fails through:

- deferral (“no clear harm yet”),
- minimization (“within acceptable bounds”),
- or diffusion (“no single decision caused this”).

Stress-testing ethical ambiguity examines whether:

- ethical concerns can trigger containment without definitive proof,
- governance allows for precautionary intervention,
- and responsibility is acknowledged despite uncertainty.

This stress category is critical because many real-world harms emerge **gradually**, not explosively.

2.7 Compounded Stress Conditions

When multiple stressors interact

In practice, governance rarely encounters a single stress type in isolation.

More often, failures occur when:

- authority stress combines with incentive pressure,
- delegation stress amplifies telemetry gaps,
- ethical ambiguity coincides with commercial urgency.

Version D therefore treats stress-testing as **compositional**, allowing scenarios where multiple stress categories interact.

Compounded stress tests examine whether governance:

- degrades predictably rather than catastrophically,
- prioritizes containment over convenience,
- and preserves accountability even when signals conflict.

Compounded stress distinguishes **robust governance** from brittle compliance.

2.8 What These Stress Types Are Not

To avoid misinterpretation, governance stress categories are **not**:

- threat models,
- vulnerability classes,
- risk taxonomies,
- or ethical score dimensions.

They do not define how “good” or “bad” a system is.

They define **where governance is likely to fracture** if not designed, exercised, and reviewed deliberately.

2.9 Chapter Summary

This chapter defines the primary types of governance stress that Version D addresses:

- authority stress,
- delegation stress,
- telemetry stress,
- incentive stress,
- ethical ambiguity stress,

- and compounded stress conditions.

Each category targets a different failure mode in governance reasoning, not system performance.

By categorizing stress explicitly, Version D enables governance stress-testing to be:

- systematic rather than ad hoc,
- comparable across systems and sectors,
- and focused on strengthening controls before harm occurs.

Chapter 3 — Governance Stress-Testing Methods (Non-Exploitative)

3.1 Why Stress-Testing Methods Must Be Constrained

Stress-testing governance introduces a paradox:

- If stress is applied too lightly, failures remain hidden.
- If stress is applied too aggressively, testing itself becomes unsafe, unethical, or misleading.

Version D therefore requires **explicitly bounded stress-testing methods**. These methods are designed to:

- reveal governance weaknesses,
- preserve system safety,
- protect third-party interests, and
- avoid creating new risk during evaluation.

Stress-testing governance is not an adversarial contest. It is a **controlled inquiry into decision behavior under pressure**.

3.2 Principles for Governance Stress-Testing

All stress-testing activities under Version D shall conform to the following principles:

1. **Non-exploitative by design**
   Stress-testing shall not involve vulnerability exploitation, security bypass, or unauthorized access.
2. **Governance-first focus**
   The object of stress is governance behavior, not model internals or performance characteristics.
3. **Human-centred oversight**
   All stress scenarios must remain reviewable, interruptible, and attributable to human decision authority.
4. **Reversible impact**
   Stress-testing shall not cause irreversible system changes, data corruption, or real-world harm.
5. **Evidence-generating, not judgment-issuing**
   Stress-testing produces observations and evidence, not approvals, certifications, or ethical scores.

These principles distinguish governance stress-testing from adversarial testing or operational audits.

3.3 Simulation-Based Stress Scenarios (SAPP-Anchored)

Simulation AI-Powered Policies (SAPP), introduced in Version A, provide the primary mechanism for **pre-impact stress testing**.

Under Version D, SAPP is used to simulate conditions that:

- increase uncertainty,
- challenge governance assumptions,
- or expose boundary ambiguity.

Examples include:

- simulated policy conflicts,
- adversarial but non-exploitative user behavior,
- escalation saturation scenarios,
- delegation chains under constrained oversight.

Simulation allows governance to be stressed **without exposing real users, markets, or infrastructure**.

Importantly, simulation outcomes are treated as **signals**, not predictions. Their purpose is to reveal how governance *responds*, not to forecast harm.

In regulatory and supervisory contexts, this class of simulation aligns with emerging regtech approaches that test governance behaviour and decision pathways before live system exposure, rather than relying on post-incident reporting or retrospective audits.

3.4 Authority Boundary Stress Exercises

Authority stress is applied through **decision-ownership challenges**, not technical manipulation.

Typical exercises include:

- scenarios where escalation authority is unclear,
- simulated time-critical decisions with incomplete evidence,
- conflicts between internal roles with overlapping responsibility.

Stress-testing authority examines whether:

- governance authority is explicit,
- decision rights are exercised rather than deferred,
- and accountability remains traceable under pressure.

No exercise may permit governance authority to be silently reassigned or bypassed during testing.

3.5 Delegation and Composition Stress Exercises

Delegation stress is applied by **composing governance scenarios**, not by breaking integrations.

Exercises may include:

- simulated delegation to third-party models under degraded telemetry,
- chain-of-delegation scenarios with ambiguous responsibility,
- withdrawal or partial failure of external services.

Stress-testing delegation examines whether governance:

- remains external to execution substrates,
- preserves accountability across composition,
- and constrains assurance when visibility degrades.

Delegation stress does not require interacting with real third-party systems. It is exercised through declared dependency models and simulated outcomes.

3.6 Telemetry Degradation and Evidence Stress

Telemetry stress is applied by **intentionally constraining evidence quality** within declared limits.

Examples include:

- delayed telemetry availability,
- partial signal loss,
- conflicting evidence streams,
- ambiguous attribution.

Stress-testing telemetry examines whether:

- governance decisions degrade conservatively,
- assurance signals are constrained rather than asserted,
- escalation pathways activate when evidence confidence falls.

Under Version D, **absence of evidence must never default to confidence**. Stress-testing verifies that this principle is operational, not rhetorical.

3.7 Ethical Boundary Walkthroughs

Ethical stress-testing is conducted through **structured walkthroughs**, not moral scoring.

These walkthroughs involve:

- plausible but ambiguous harm scenarios,
- competing ethical priorities,
- uncertainty about affected parties.

Participants are asked to articulate:

- what decision they would take,
- what evidence they would require,
- where they would pause or constrain operation,
- how they would document responsibility.

The objective is not consensus. It is to surface **where ethical authority resides and how it is exercised under ambiguity**.

3.8 Escalation and Revocation Rehearsals

A governance system that cannot rehearse escalation cannot be trusted to execute it.

Under Version D, stress-testing includes **revocation rehearsal**, such as:

- simulated downgrading of assurance signals,
- temporary suspension of system capability,
- invocation of human-in-the-loop overrides.

These rehearsals examine whether:

- revocation is technically and procedurally possible,
- organizational resistance emerges,
- accountability is preserved during rollback.

Revocation rehearsal treats degradation as a **normal governance action**, not an exceptional failure.

3.9 What Stress-Testing Must Explicitly Avoid

To preserve safety and standards credibility, Version D stress-testing shall **not** include:

- security exploitation or penetration testing,
- model jailbreaks or prompt hacking,
- disclosure of sensitive attack vectors,
- public ranking or ethical scoring,
- production-environment experimentation without containment.

These activities belong to other domains and must not be conflated with governance stress-testing.

3.10 Outputs of Governance Stress-Testing

Stress-testing produces **governance evidence**, including:

- identified failure modes,
- clarified authority boundaries,
- revised escalation thresholds,
- constrained assurance conditions,
- documented ethical decision rationales.

It does **not** produce:

- approval statements,
- alignment claims,
- certification artifacts,
- or deployment authorization.

Outputs feed back into:

- policy refinement,

- containment strengthening,
- assurance interpretation limits.

3.11 Chapter Summary

This chapter defines **how governance is stressed safely**:

- through simulation, not exploitation;
- through decision challenges, not system compromise;
- through evidence degradation, not evidence fabrication;
- through ethical ambiguity, not moral abstraction.

By constraining stress-testing methods explicitly, Version D ensures that governance can be exercised under pressure **without creating new risk or false confidence**.

Chapter 4 — Ethics Under Governance Stress

4.1 Why Ethics Emerges Only Under Pressure

Ethical intent is often articulated at design time, when choices are abstract and consequences are distant. Under these conditions, ethical positions tend to converge and appear uncontroversial.

Ethical reality, however, emerges **under pressure**, when decisions must be made with incomplete information, competing obligations, and tangible consequences.

Version D therefore treats ethics not as a static alignment state, but as a **governance function exercised under stress**. Ethics is revealed by how decisions are made when:

- evidence is ambiguous,
- harm is plausible but not yet realized,
- intervention carries cost or delay, and
- responsibility cannot be deferred to procedure.

Ethics that cannot be exercised under these conditions is not operational ethics.

4.2 Ethics as a Governance Capability, Not a Principle Set

Many ethical frameworks emphasize principles such as fairness, accountability, transparency, and human oversight. While these principles are necessary, they do not determine **who decides**, **when to intervene**, or **how to act under uncertainty**.

Within Version D:

- ethics is treated as a **decision capability**,
- exercised through governance controls,
- bounded by authority and accountability,
- and evidenced through action and restraint.

Ethics is therefore inseparable from governance structure. Where governance authority is unclear, ethical responsibility becomes diffuse. Where escalation is blocked, ethical intent becomes performative.

4.3 Ethical Authority and Non-Delegability

A central ethical question in AI governance is **whether ethical judgment can be delegated**.

Version D asserts that ethical judgment is **non-delegable**, even when AI systems assist with analysis, simulation, or drafting.

Accordingly:

- AI may surface ethical considerations but shall not resolve them.
- Ethical trade-offs shall not be automated.
- Responsibility for ethical decisions shall remain human, attributable, and defensible.

Stress-testing ethics therefore examines whether governance mechanisms **preserve ethical authority** under pressure, rather than whether ethical principles are declared.

4.4 Ethical Decision-Making Under Uncertainty

Ethical failure often occurs not through overt harm, but through **delay, minimization, or normalization** in the face of uncertainty.

Common ethical stress conditions include:

- uncertainty about affected populations,
- indirect or long-term impacts,
- competing interpretations of acceptable risk,
- lack of definitive evidence.

Version D does not require ethical certainty. It requires **ethical containment**.

Ethical containment means:

- the ability to pause or constrain operation,
- explicit acknowledgment of uncertainty,
- documentation of decision rationale,
- willingness to escalate without proof of harm.

Under Version D, **ethical restraint is a valid and expected outcome**.

4.5 The Ethics of "Not Deciding"

One of the most under-examined ethical acts in AI governance is the decision **not to decide**.

Non-decision may take the form of:

- deferring deployment,
- limiting scope,
- suspending a capability,
- requiring additional evidence before proceeding.

Version D treats non-decision not as failure, but as a **first-class ethical action** when conditions do not justify progression.

Stress-testing ethics therefore includes scenarios where the ethically defensible action is to:

- halt,
- slow,
- or reverse.

Governance systems that cannot support ethical non-decision are structurally biased toward action - even when action is unjustified.

4.6 Ethical Escalation and Accountability

Ethical decisions under stress require **escalation pathways** that are:

- explicit,
- accessible,
- and empowered.

Version D examines whether ethical concerns can:

- trigger escalation without stigma,
- reach decision-makers with authority,
- and result in proportional intervention.

Ethical escalation is not whistleblowing. It is a **designed governance pathway**.

Stress-testing ethics reveals whether ethical escalation is operational, or merely aspirational.

4.7 Documenting Ethical Reasoning as Evidence

Ethical decisions are often defended retrospectively through narrative. Version D requires that ethical reasoning be **documented contemporaneously** as governance evidence.

Such documentation should include:

- the decision context,
- the uncertainties acknowledged,
- the alternatives considered,
- the rationale for action or restraint,
- the accountable decision-maker.

This documentation does not serve to justify decisions after the fact. It serves to **make ethical reasoning inspectable**, contestable, and improvable.

Ethics without evidence is indistinguishable from opinion.

4.8 Avoiding Ethical Theatre

Ethical theatre occurs when:

- ethics committees exist without authority,
- principles are published without enforcement,
- impact assessments are completed without consequence,
- or ethical language substitutes for ethical action.

Version D explicitly rejects ethical theatre.

Ethical governance is defined by **what changes when ethics are invoked**, not by how ethics are described.

Stress-testing ethics surfaces whether ethical concerns can meaningfully alter system behavior, governance posture, or assurance interpretation.

4.9 Ethics, Trust, and Revocation

Ethical failure does not always require harm to have occurred. It may arise when:

- ethical boundaries are knowingly eroded,

- uncertainty is ignored,
- or governance constraints are bypassed.

Under Version D, ethical stress-testing may justify:

- constraining assurance signals,
- downgrading trust posture,
- or rehearsing revocation.

Ethics therefore interacts directly with trust and assurance, not as a separate domain but as a **governance signal**.

Trust that ignores ethical stress is incomplete.

Ethical frameworks and principles such as fairness, accountability, transparency, and human oversight do not gain operational traction through downstream, after-the-fact assessment alone. In practice, ethical confidence emerges upstream, where governance mechanisms shape decisions before harm, ambiguity, or escalation occurs.

In this sense, ethics is inseparable from experience. Repeated, evidenced ethical restraint under stress builds confidence that governance can be relied upon when conditions degrade. It is this accumulated, experience-based signal - rather than retrospective ethical evaluation - that informs higher-order trust and assurance interpretations, including the maturity and stability of upstream governance signals.

4.10 Chapter Summary

This chapter reframes ethics as a **governance capability exercised under stress**, not a design-time declaration.

Key points:

- Ethics is revealed through decisions under uncertainty.
- Ethical authority is non-delegable.
- Ethical restraint and non-decision are valid outcomes.
- Ethical escalation must be operational, not symbolic.
- Ethical reasoning must be documented as evidence.
- Ethics interacts directly with trust, assurance, and revocation.

Version D therefore treats ethical evaluation as inseparable from governance stress-testing, ensuring that ethical claims are grounded in **decision behavior**, not intent statements.

Chapter 5 — Evidence Produced by Governance Stress-Testing

5.1 Why Stress-Testing Evidence Must Be Distinguished

Governance stress-testing generates evidence that is **qualitatively different** from:

- performance benchmarks,
- security findings,
- compliance artifacts, or
- assurance certifications.

If this distinction is not made explicit, stress-testing risks being misused to:

- imply approval,
- justify deployment,
- or replace lifecycle assurance.

Version D therefore defines **stress-testing evidence** as a specific class of governance evidence, with **clear scope, limits, and interpretation rules**.

5.2 What Counts as Governance Stress-Testing Evidence

Under Version D, governance stress-testing produces evidence about **governance behavior**, not system correctness.

Valid evidence types include:

- **Observed failure modes**
  Where authority, escalation, or evidence handling degraded under stress.
- **Boundary clarifications**
  Explicit identification of where governance assumptions broke or became ambiguous.
- **Decision rationales under pressure**
  Documented reasoning for action, restraint, escalation, or non-decision.
- **Control responsiveness**
  Whether governance mechanisms activated as designed when stressed.
- **Ethical constraint articulation**
  How ethical uncertainty was handled, constrained, or escalated.

This evidence reflects **how governance behaves**, not whether a system is “safe” or “aligned.”

5.3 What Stress-Testing Evidence Is Not

To prevent misuse, Version D explicitly excludes the following interpretations:

Stress-testing evidence is **not**:

- proof of safety or alignment,
- evidence of compliance,
- justification for certification,
- validation of readiness,
- or endorsement of deployment.

Stress-testing outcomes may include "successful" responses under stress, but such outcomes **do not elevate trust status**. They only inform whether governance mechanisms are **credible under pressure**.

5.4 Evidence Quality and Documentation Requirements

Stress-testing evidence must meet minimum quality standards to be meaningful:

- **Contextualized**
  The stress condition, assumptions, and constraints must be documented.
- **Attributable**
  Decisions must be traceable to accountable human roles.
- **Time-bound**
  Evidence reflects behavior under specific conditions, not general capability.
- **Non-generalized**
  Findings must not be extrapolated beyond tested scope.

Version D discourages narrative summaries without traceable decision context. Where documentation is incomplete, evidence shall be treated as **informative but non-reliable**.

5.5 Relationship to CSC and Assurance Signals

Stress-testing evidence interacts with assurance **only by constraint**, never by elevation.

Accordingly:

- Stress-testing does **not increase** CSC levels.
- Stress-testing outcomes may **cap**, **qualify**, or **constrain** assurance.

- Unresolved stress-test failures may justify:
    - provisional assurance,
    - reduced scope,
    - or revocation readiness.

If stress-testing reveals governance weaknesses, assurance claims must be **narrowed**, not defended.

5.6 Using Stress-Testing Evidence for Governance Improvement

The primary use of stress-testing evidence is **governance refinement**, including:

- tightening escalation thresholds,
- clarifying authority boundaries,
- revising policy constraints,
- improving telemetry completeness,
- strengthening ethical intervention pathways.

Stress-testing evidence supports **learning and correction**, not validation.

Repeated stress-testing over time may demonstrate **governance maturation**, but maturity remains a narrative assessment supported by evidence, not a scored outcome.

5.7 Evidence Retention and Disclosure Boundaries

Stress-testing evidence often contains sensitive material, including:

- hypothetical failure scenarios,
- internal decision weaknesses,
- governance design gaps.

Version D requires **controlled handling** of such evidence:

- Evidence may be retained internally for governance learning.
- Disclosure to third parties shall be:
    - proportional,
    - contextualized,
    - and non-exploitative.
- Public disclosure of stress-testing detail is **not required** and may be inappropriate.

Absence of public disclosure does not invalidate governance learning. Transparency must be balanced with safety.

5.8 Avoiding "Stress-Testing Theatre"

Stress-testing becomes performative when:

- scenarios are chosen to confirm success,
- failures are reframed as acceptable,
- documentation replaces corrective action,
- or results are used for signaling rather than improvement.

Version D explicitly treats **suppression or misrepresentation of stress-testing evidence** as a governance failure.

If governance cannot acknowledge its own stress-induced weaknesses, it cannot claim credibility.

5.9 Chapter Summary

This chapter defines how governance stress-testing evidence is:

- generated,
- constrained,
- interpreted,
- and used.

Key principles:

- Stress-testing evidence reveals governance behavior, not system fitness.
- Evidence constrains assurance; it does not elevate it.
- Ethical and authority failures are valid, expected findings.
- Evidence must inform governance refinement, not justification.
- Suppression or misuse of findings undermines trust.

Stress-testing evidence exists to **remove false confidence**, not to create new claims.

Chapter 6 — Relationship to CSC, Trust, and Revocation

6.1 Why Stress-Testing Must Constrain Trust, Not Inflate It

Governance stress-testing is often misinterpreted as a way to *increase* confidence.
Version D explicitly rejects this framing.

Stress-testing exists to **challenge trust assumptions**, not to validate them.
A governance system that performs well only under favorable conditions is not trustworthy, it is untested.

Accordingly, stress-testing evidence may **restrict**, **qualify**, or **delay** trust signals, but it shall never be used to elevate trust status or justify deployment readiness.

6.2 CSC Authority Remains Defined Exclusively by Version A

The Compliance Star Certificate (CSC) is defined entirely by Version A:

- assurance semantics,
- scoring logic,
- evidence requirements,
- degradation and revocation rules.

Version D introduces **no new CSC criteria**, weights, star levels, or scoring pathways.

Stress-testing outcomes do **not**:

- increase CSC scores,
- substitute for telemetry,
- override CSC qualification gates,
- or justify star-level progression.

CSC authority remains unchanged and external to stress-testing activities.

6.3 How Stress-Testing Interacts with CSC (Constraint Path)

Stress-testing affects CSC only through **constraint**, not computation.

Valid interactions include:

- **Scope limitation**
  Stress-testing may reveal that assurance applies only to a narrower context than previously assumed.
- **Provisional qualification**
  Unresolved stress-test failures may justify marking CSC outputs as provisional or exploratory.
- **Degradation readiness**
  Stress-testing may identify conditions under which CSC degradation or revocation must be triggered rapidly.

Stress-testing therefore functions as a **trust boundary clarifier**, not an assurance generator.

6.4 Stress-Testing as a Precursor to Revocation Readiness

Version A defines revocation as a protective governance action, not a punitive one.
Version D ensures that revocation is **operationally realistic**.

Stress-testing examines whether:

- revocation pathways exist,
- revocation can be executed without institutional resistance,
- trust signals can be degraded without narrative manipulation,
- and accountability remains intact during rollback.

A governance system that cannot rehearse revocation cannot credibly claim revocation readiness.

6.5 Trust Degradation as a Valid Outcome

Under Version D, trust degradation is not a failure condition - it is an **expected governance outcome**.

Stress-testing may legitimately result in:

- constrained assurance,
- reduced CSC scope,
- downgraded star levels,
- or temporary suspension of trust signals.

Attempting to preserve trust signals in the face of unresolved stress-test findings constitutes governance failure.

Trust that cannot degrade cannot be trusted.

6.6 Stress-Testing and Lifecycle Governance

Stress-testing evidence is **time-bound** and **context-specific**.

Accordingly:

- successful stress-testing does not confer permanent trust;
- new stress conditions may invalidate prior findings;
- system evolution may require re-entry into stress-testing.

Version D therefore treats stress-testing as a **recurring lifecycle activity**, not a one-time gate.

Stress-testing supports continuous governance by ensuring that trust claims remain **conditioned on current behavior**, not historical success.

6.7 Preventing "Stress-Test-as-Certification"

A critical guardrail in Version D is the explicit prevention of stress-testing being misrepresented as certification.

Accordingly:

- stress-testing results shall not be marketed,
- stress-testing completion shall not be represented as approval,
- absence of observed failure shall not be equated with alignment.

Any representation of stress-testing as endorsement, readiness, or certification is a governance breach.

6.8 Accountability During Trust Constraint and Revocation

Stress-testing often surfaces uncomfortable governance truths.

Version D requires that:

- trust constraint decisions remain attributable to human authority,
- justification for constraint or revocation is documented,
- organizational pressure does not override evidence.

Accountability does not end when trust is reduced—it intensifies.

6.9 Chapter Summary

This chapter clarifies how Version D integrates with trust and assurance mechanisms defined in Version A.

Key points:

- Stress-testing constrains trust; it does not elevate it.
- CSC authority and semantics remain unchanged.
- Stress-testing supports revocation readiness.
- Trust degradation is a valid and expected outcome.
- Stress-testing is lifecycle-bound, not a one-time gate.
- Misrepresentation of stress-testing as certification is prohibited.

Governance credibility is not established by passing stress-tests. It is established by **responding honestly when stress exposes weakness**.

Chapter 7 — Who Should Use Version D, and When

7.1 Why Audience and Timing Matter

Governance stress-testing is powerful precisely because it can surface uncomfortable truths. Used at the wrong time, or by the wrong actors, it can also be misapplied, misunderstood, or weaponized.

Version D therefore defines **who should use governance stress-testing**, **under what conditions**, and **for what purpose**. This is not an access control mechanism; it is a **use-discipline requirement** intended to preserve governance integrity and standards credibility.

7.2 Intended Primary Users

Version D is intended for actors who hold, influence, or evaluate **governance authority**, not for those seeking validation, benchmarking, or market signaling.

Regulators and Oversight Bodies

Regulators may use Version D to:

- evaluate whether governance mechanisms remain credible under stress,
- assess revocation readiness and escalation realism,
- identify systemic governance blind spots before harm occurs.

Version D does not provide regulatory approval criteria. It supports **oversight literacy**, not enforcement thresholds.

Standards Bodies and Technical Committees

Standards organizations may use Version D to:

- explore how governance frameworks behave under adverse conditions,
- assess whether proposed standards support evidence-based trust under stress,
- identify where standards language may fail to anticipate real-world governance pressure.

Version D is suitable for **standards exploration and stress-informed drafting**, not for normative requirement setting.

Organizations Operating or Scaling AI Systems

Organizations may use Version D to:

- stress-test governance before scale-up or integration,
- evaluate whether escalation and revocation mechanisms are operationally realistic,
- surface ethical and authority failure modes prior to real-world exposure.

Version D should be applied **before confidence is asserted**, not after incidents occur.

Boards, Risk Committees, and Executive Oversight

Governance stress-testing is particularly relevant for decision-makers who must:

- sign off on risk posture,
- accept accountability for trust claims,
- authorize expansion into new contexts or markets.

For these actors, Version D provides a structured way to ask:

"What happens to governance when things go wrong?"

7.3 When Version D Should Be Applied

Version D is most appropriate at **decision inflection points**, including:

- prior to deployment into higher-impact contexts,
- before expanding scope, autonomy, or delegation,
- when onboarding new models, tools, or data sources,
- following material system changes,
- when assurance signals appear stable but untested.

It is **not** intended for continuous application during steady-state operation.
It is a **deliberate stress activity**, not an always-on monitoring function.

7.4 When Version D Should Not Be Applied

To prevent misuse, Version D should **not** be applied:

- as a public benchmarking exercise,
- as a prerequisite for market access,
- as a substitute for regulatory compliance,
- as a red-team security activity,
- as a marketing or signaling tool.

Applying Version D in these contexts risks converting governance stress-testing into **governance theatre** or coercive evaluation, undermining its purpose.

7.5 Relationship to Versions A, B, and C

Version D does not stand alone.

Its proper use depends on the existence of:

- **Version A** — governance authority, assurance semantics, and CSC logic,
- **Version B** — execution-level feasibility and telemetry realism (where applicable),
- **Version C** — disciplined onboarding and containment.

Version D assumes that these foundations exist.
It does not compensate for their absence.

Stress-testing governance without authority, telemetry, or containment is not stress-testing - it is speculation.

7.6 Relationship to the v1.0-workshop

The **v1.0-workshop** is a prerequisite for meaningful use of Version D.

The workshop develops:

- governance reasoning discipline,
- accountability ownership,
- boundary articulation,
- comfort with uncertainty and non-decision.

Version D assumes participants can already:

- articulate intent and scope clearly,
- identify non-delegable judgment,
- reason under ambiguity,
- document defensible decisions.

Without this human capability substrate, Version D risks being misread as abstract ethics discussion or adversarial testing. With it, Version D becomes **practical and consequential**.

Informative reference:

Additional information (referenced in second edition notes page 4) about the Designing Governance-Ready AI Outputs workshop, including an overview and participant information pack, is available at:

https://policyofdevelopments.org/workshop

7.7 Ethical Preconditions for Using Version D

Because stress-testing may surface sensitive governance weaknesses, Version D requires:

- declared purpose and scope,
- defined ethical rules of engagement,
- protection against misuse or misrepresentation,
- clear accountability for findings.

Version D should never be used to:

- entrap participants,
- assign blame retroactively,
- or justify predetermined outcomes.

Ethical use of Version D is itself a governance test.

7.8 Chapter Summary

This chapter clarifies that Version D is:

- for regulators, standards bodies, operators, and accountable decision-makers;

- to be used at governance inflection points, not continuously;
- dependent on Versions A–C and the v1.0-workshop;
- unsuitable for certification, benchmarking, or marketing;
- ethically bounded by declared purpose and accountability.

Governance stress-testing is not a universal tool.
It is a **deliberate, high-responsibility activity** intended to strengthen trust by confronting fragility before it becomes harm.

Closing Statement

Trust, Stress, and Governance Integrity

This document has introduced governance stress-testing and ethical evaluation as **core governance capabilities**, not supplementary exercises. Its purpose has not been to prescribe new controls, define compliance criteria, or assert trust, but to expose where governance assumptions fail when conditions become uncertain, pressured, or ethically ambiguous.

Version D reinforces a central principle of this platform: **trust cannot be declared, certified, or inferred from design intent alone**. Trust emerges only when governance mechanisms remain effective under stress - when authority is exercised rather than deferred, when evidence constrains decisions rather than justifies them, and when ethical responsibility is preserved despite uncertainty.

Stress-testing governance is therefore not about proving robustness. It is about **removing false confidence**. Failure modes identified through stress are not defects to be concealed; they are governance evidence that enables correction, restraint, and maturation before harm occurs. A governance system that cannot tolerate stress without collapse cannot credibly claim trustworthiness at scale.

Ethics, within this framework, has been treated not as a checklist or alignment label, but as a **decision practice under constraint**. Ethical governance is demonstrated through escalation, non-decision, revocation readiness, and documented accountability when evidence is incomplete or contested. Ethics that cannot alter system behavior under pressure is not operational ethics.

Version D remains explicitly subordinate to the governance authority, assurance semantics, and conformance meaning defined in **Version A**, complementary to the execution-level feasibility explored in **Version B**, and dependent on the onboarding and containment discipline formalized in **Version C**. It introduces no new certification logic, scoring mechanisms, or approval pathways.

Effective use of this document presumes the **human governance capability** developed through the *Designing Governance-Ready AI Outputs* workshop. Without disciplined professional reasoning, stress-testing risks becoming abstract or performative. With it, stress-testing becomes a practical mechanism for accountable decision-making under uncertainty.

Taken together, Versions A–D and the workshop articulate a coherent governance posture:

- trust is engineered before impact,
- observed during operation,
- constrained under stress, and
- withdrawn when evidence no longer supports confidence.

Governance integrity is not demonstrated by the absence of failure, but by the **willingness and ability to respond when stress exposes fragility**. This document exists to support that response - before harm occurs.

The practical test of this framework is not whether it can be explained, but whether governance actors can act with restraint, accountability, and ethical clarity as systems and organizations operate under differing conditions.